

Commerce XI Maths App

Theory (Pages 40–42)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 22

Question 1 - MCQ · PG-40 · TH-Q1

If x is a cube root of unity, then

Options

A. $x^3 = 1$

B. $x^2 = 1$

C. $x^3 = 0$

D. $x = -1$ only

Correct answer: $x^3 = 1$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $x^3 = 1$.

Question 2 - MCQ · PG-40 · TH-Q2

Factor $x^3 - 1$

Options

A. $(x - 1)(x^2 + x + 1)$

B. $(x + 1)(x^2 - x + 1)$

C. $(x - 1)^3$

D. $(x - 1)(x^2 - x - 1)$

Correct answer: $(x - 1)(x^2 + x + 1)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(x - 1)(x^2 + x + 1)$.

Question 3 - MCQ · PG-40 · TH-Q3

The three cube roots of unity are

Options

A. $1, (-1 + \sqrt{3})i/2, (-1 - \sqrt{3})i/2$

B. $1, i, -i$

C. $1, -1, i$

D. $1, (1 + \sqrt{3})i/2, (1 - \sqrt{3})i/2$

Correct answer: $1, (-1 + \sqrt{3})i/2, (-1 - \sqrt{3})i/2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $1, (-1 + \sqrt{3})i/2, (-1 - \sqrt{3})i/2$.

Question 4 - MCQ · PG-40–41 · TH-Q4

If $w = (-1 + \sqrt{3}i)/(2)$, then w^2 equals

Options

- A. $(-1 - \sqrt{3}i)/(2)$
- B. $(-1 + \sqrt{3}i)/(2)$
- C. $(1 - \sqrt{3}i)/(2)$
- D. 1

Correct answer: $(-1 - \sqrt{3}i)/(2)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(-1 - \sqrt{3}i)/(2)$.

Question 5 - MCQ · PG-41 · TH-Q5

For a complex cube root of unity w , w^3 equals

Options

- A. 1
- B. 0
- C. -1
- D. w

Correct answer: 1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 1.

Question 6 - MCQ · PG-41 · TH-Q6

For $w \neq 1$, which relation holds?

Options

- A. $w^2 + w + 1 = 0$
- B. $w^2 - w + 1 = 0$
- C. $w^2 + w - 1 = 0$
- D. $w^2 = 1$

Correct answer: $w^2 + w + 1 = 0$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $w^2 + w + 1 = 0$.

Question 7 - MCQ · PG-41 · TH-Q7

From $w^2 + w + 1 = 0$, $w + w^2$ equals

Options

- A. -1
- B. 1
- C. 0
- D. w

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1 .

Question 8 - MCQ · PG-41 · TH-Q8

For a complex cube root of unity w , $(1)/(w)$ equals

Options

- A. w^2
- B. w
- C. 1
- D. $-w$

Correct answer: w^2

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w^2 .

Question 9 - MCQ · PG-41 · TH-Q9

For a complex cube root of unity w , $(1)/(w^2)$ equals

Options

- A. w
- B. w^2
- C. 1
- D. $-w^2$

Correct answer: w

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w .

Question 10 - MCQ · PG-41 · TH-Q10

For $n \in \mathbb{N}$, w^{3n} equals

Options

- A. 1
- B. w
- C. w^2
- D. 0

Correct answer: 1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 1.

Question 11 - MCQ · PG-41 · TH-Q11

For $n \in \mathbb{N}$, w^{3n+1} equals

Options

- A. w
- B. 1
- C. w^2
- D. $-w$

Correct answer: w

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w .

Question 12 - MCQ · PG-41 · TH-Q12

For $n \in \mathbb{N}$, w^{3n+2} equals

Options

- A. w^2
- B. 1
- C. w
- D. $-w^2$

Correct answer: w^2

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w^2 .

Question 13 - MCQ · PG-41 · TH-Q13

The conjugate of w is

Options

- A. w^2
- B. w
- C. 1
- D. $-w$

Correct answer: w^2

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w^2 .

Question 14 - MCQ · PG-41 · TH-Q14

The conjugate of w^2 is

Options

- A. w
- B. w^2
- C. 1
- D. $-w^2$

Correct answer: w

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w .

Question 15 - MCQ · PG-41 · TH-Q15

Find $(1)/(w) + (1)/(w^2)$

Options

- A. -1
- B. 1
- C. 0
- D. -2

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1 .

Question 16 - MCQ · PG-41 · TH-Q16

Find $(1 + w^2)^3$

Options

- A. -1
- B. 1
- C. 0
- D. -8

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1.

Question 17 - MCQ · PG-41 · TH-Q17

Find $(1 - w + w^2)^3$

Options

- A. -8
- B. 8
- C. -1
- D. 0

Correct answer: -8

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -8.

Question 18 - MCQ · PG-41 · TH-Q18

Find $(1 - w)(1 - w^2)(1 - w^4)(1 - w^5)$

Options

- A. 9
- B. 3
- C. -9
- D. 0

Correct answer: 9

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 9.

Question 19 - MCQ · PG-41–42 · TH-Q19

If n is a multiple of 3, then $1 + w^n + w^{2n}$ equals

Options

- A. 3
- B. 0
- C. 1
- D. -1

Correct answer: 3

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 3.

Question 20 - MCQ · PG-41–42 · TH-Q20

If n is not a multiple of 3, then $1 + w^n + w^{2n}$ equals

Options

- A. 0
- B. 3
- C. 1
- D. -1

Correct answer: 0

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 0.

Question 21 - MCQ · PG-40 · TH-Q21

Among the cube roots of unity, the two non-real roots are

Options

- A. complex conjugates of each other
- B. equal real numbers
- C. additive inverses only
- D. both equal to i

Correct answer: complex conjugates of each other

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives complex conjugates of each other.

Question 22 - MCQ · PG-41 · TH-Q22

Which notation represents the cube roots of unity?

Options

- A. $1, w, w^2$
- B. $0, w, w^2$
- C. $1, w, w^3$
- D. $-1, w, w^2$

Correct answer: $1, w, w^2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $1, w, w^2$.